INCH - POUND

MIL-L-45194G
27 April 1989
SUPERSEDING
MIL-L-45194F
1 July 1983

MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC BELT, 20MM: M14A2/M14A2N

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of metallic belt links intended to hold 20 millimeter (mm) ammunition in the form of flexible belts for use in the M61 or M197 (M89EI Feeder) type 20mm automatic weapon systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-13855	- Weapons, Small Arms, General Specification for
DOD-P-16232	- Phosphate, Coating, Heavy, Manganese or Zinc Base
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-W-12332	- Welding, Resistance, Spot, Seam and Projection:for Fabricating Assemblies of Low Carbon Steel (Navy Use Only)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1168 - Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

8447633 - Link, Cartridge, Metallic Belt, 20mm: M14A2
P8447633 - Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20mm: M14A2
8447634 - Loop, Long
8447635 - Loop, Short
8447859 - Test Diagram and Requirements for Link, Cartridge, Metallic Belt, 20mm: M14A2 or M14A2N
#12910231 - Link, Cartridge, Metallic Belt, 20mm: M14A2N
#P12910231 - Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20mm: M14A2N
#12910232 - Loop, Long with Shield for Link M14A2N
#12910233 - Shield, Radhaz for Link M14A2N

(# NAVY USE ONLY)

INSPECTION EQUIPMENT DRAWINGS

42845938 - Radhaz Shield Stand Off Go-No Go Gage
7190390 - Width Gage
7190394 - Progressive Snap Gage
7190398 - Flush Pin Gage
7190406 - Depth Gage
7190456 - Progressive Snap Gage
7190457 - Depth Gage
7190458 - Width Gage

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7190471	- Pull Test Fixture Gage
7190482	- Width Gage
7274001	- Gage, Fixture, Test
8409005	- Loading Fixture
84409191	- Gage, Dial Indicating
11838231	- Fixture, Test, Extraction
11838232	- Pump Assembly, Hydraulic, Electric Powered
11838277	- Set-up and Operating Instructions

(# NAVY USE ONLY)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

ASTM B117	- Standard Method of Salt Spray (fog) Testing
ASTM E18	- Method of Test of Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components. The links shall comply with all requirements specified on drawings 8447633, 12910231 and 8447859 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Short loop displacement. The short loop shall be capable of hand displacement with respect to the long loop and upon release shall return to its own original position by spring action.

3.4 Free hinging. Links assembled into belts of 100 links each using M51 series 20MM dummy cartridges shall hinge and fold over without binding, kinking, separating or any interference. The test shall be performed as specified in 4.5.7.

3.5 Functioning. The M14A2 links shall be capable of being assembled into belts with 20mm cartridges. The belts will be utilized in any or all of the belt fed or linkless M61 Gun Systems which are designed for M14 links. The links shall not crack, break, separate or cause a malfunction. The test shall be performed as specified in paragraph 4.5.8.

3.5.1 The M14A2N links shall be capable of being assembled into belts with 20mm cartridges. The belts shall be cycled in the M197 Gun System with the M89E1 Feeder. The links shall not crack, break, separate or cause a malfunction. The test shall be performed as specified in paragraph 4.5.8.1.

3.6 Marking. The M14A2 links shall be marked in accordance with Drawing (Dwg) 8447634. In addition, the links shall be identified with the inspection lot number as specified in MIL-STD-1168.

3.6.1 The M14A2N links shall be marked in accordance with Drawings (Dwg.) 8447634 and 12910232. In addition, the links shall be identified with the inspection lot number as specified in MIL-STD-1168.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. Workmanship shall be in accordance with MIL-W-13855. Cracks, tears and burrs in non-principal areas and deformations shall not exceed the limits specified in 6.4.1, 6.4.2.2 and 6.4.3.2.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Resposibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Link, Cartridge, Metallic Belt, 20MM: M14A2	8447633	2400
Loop, short	8447635	263
Loop, long	8447634	263
(Navy Use Only)		
Link, Cartridge, Metallic Belt, 20MM: M14A2N	12910231	2100
Loop, long with Shield	12910232	295
Shield, Radhaz	12910233	50

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. Rejection of the first article sample shall be in accordance with Table I and MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE Link, Cartridge, Metallic Belt, 20MM: M14A2		SHEET 1 OF 3		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Loop, short Dwg. 8447635)				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.1
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.2
	Salt Spray	8	ACC-0 REJ-1	3.2	4.5.3
	Hardness	50	ACC-0 REJ-1	3.2	4.5.4
	Loop, long (Dwg. 8447634)				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.2
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.2
	Salt Spray	8	ACC-0 REJ-1	3.2	4.5.3
	Hardness	50	ACC-0 REJ-1	3.2	4.5.4

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-L-45194G

PARAGRAPH	TITLE: Link, Cartridge, Metallic Belt, 20MM: M14A2 or M14A2N		SHEET 2 OF 3		DRAWING NUMBER: See Below; NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Link, Cartridge, Metallic Belt, 20mm: M14A2 or M14A2N (Dwg. 8447633 or 12910231				
	Functioning				
	M14A2 Link	600	ACC-0 REJ-1	3.5	4.5.8
	M14A2N Link	300	ACC-0 REJ-1	3.5.1	4.5.8.1
	Test Diagrams and Requirements (for Link, Cartridge, Metallic Belt, 20MM, M14A2 or M14A2N) (Dwg. 8447859)				
	Examination for defects	500	ACC-0 REJ-1	3.2	4.4.2.5 1/
	Free hinging	800	ACC-0 REJ-1	3.4	4.5.7
	Cartridge extraction	500	ACC-4 REJ-5	3.2	8409005, 11838231 11838232, 11838277, 4.5.6
	Pull and elongation	200	ACC-1 REJ-2	3.2	7190471/4.5.1 2/
	Tensile load	200	ACC-1 REJ-2	3.2	7274001, 4.5.5 2/

NOTES: 1/ Inspections shall be performed for Majors 101 and 102, concurrently with cartridge extraction test. 2/ Selected from cartridge extraction test sample.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45194G

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M14A2N (NAVY USE ONLY)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sheild, Radhaz (Dwg 12910233) Hardness	50	ACC-0 REJ-1	3.2	4.5.4
	Loop, Long with Shield Radhaz (Dwg 12910232)				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.3
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.2
	Salt Spray	8	ACC-0 REJ-1	3.2	4.5.3
	Hardness	50	ACC-0 REJ-1	3.2	4.5.4
	Weld Integrity	32	ACC-0 REJ-1	3.2	4.5.9

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and MIL-W-13855. In addition, inspection-lots shall not exceed 500,000 links.

4.4.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one charge or in a continuous type of furnace during an uninterrupted period extending over not more than one ten hour period.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, unless otherwise specified.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45194G

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Loop, Short (Link, 20MM, M14A2)	1 OF 1	8447635
			NEXT HIGHER ASSEMBLY 8447633

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:					
101.	Cracks-Tears or burrs in principal areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
102.	Short loop displacement (at assembly)		.25%	3.3	Visual/Manual
103.	Diameter, Inside		.25%	3.2	Gage
104.	Hump flange opening, min		.25%	3.2	7190458
105.	Inside width of hump, min		.25%	3.2	7190482
106.	Overall length		.25%	3.2	7190456
107.	Inside hump to inside diameter, max		.65%	3.2	7190457
108.	Thickness of flange (2)		.65%	3.2	Gage
109.	Radius (4) missing from ends of flange		.65%	3.2	Visual
	TESTS				
110.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.2
111.	Salt Spray	2/	2/	3.2	4.5.3
112.	Hardness	1/3/	.40%	3.2	4.5.4
MINOR:					
201.	Incorrect or missing protective finish		2.50%	3.2	Visual
202.	Poor workmanship		4.00%	3.8	6.4.1/6.4.2.2/6.4.3.2/ Visual

NOTES: 1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Loop, Long (Link, 20MM, M14A2)		8447634
			NEXT HIGHER ASSEMBLY 8447633

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:					
101.	Cracks-Tears or burrs in principal areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
102.	Distance from protruding end detent to side detent, Max (2)		.65%	3.2	Gage
103.	Front guide to rear locating tab		.65%	3.2	8449019
104.	Width of stripper bridge opening, min 4/		.65%	3.2	7190390
105.	Depth of guide channel, max (2)		.65%	3.2	7190398
106.	Width across nesting edge opening		.65%	3.2	Gage
107.	Height of tab or depth of undercut with respect to nesting edges, max (2)		.65%	3.2	Gage
108.	Distance from hole to outside edge of nesting area		.65%	3.2	Gage
	TESTS				
109.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.2
110.	Salt Spray	2/	2/	3.2	4.5.3
111.	Hardness	1/3/	.40%	3.2	4.5.4

NOTES:
1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3
4/ To be performed with dummy cartridge assembled to link.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 8447634
4.4.2.2	Loop, Long (Link, 20MM, M14A2)				NEXT HIGHER ASSEMBLY 8447633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR:					
201.	Height of stripper		1.50%	3.2	7190394
202.	Overall length of body, max		1.50%	3.2	Gage
203.	Depth of side detent, min (2)		1.50%	3.2	7190406
204.	Location of stripper opening		1.50%	3.2	Gage
205.	Improper or missing identification		1.50%	3.6	Visual
206.	Incorrect or missing protective finish		2.50%	3.2	Visual
207.	Poor workmanship		4.00%	3.8	Visual/6.4.1/6.4.2.2/ 6.4.3.2
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45194G

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 12910232
4.4.2.3	Loop, Long with Shield (Navy Use Only)		NEXT HIGHER ASSEMBLY 12910231

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:					
101.	Cracks-Tears or burrs in principal areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
102.	Distance from protruding end detent to side detent, Max (2)		.65%	3.2	Gage
103.	Front guide to rear locating tab		.65%	3.2	8440919
104.	Width of stripper bridge opening, min 4/		.65%	3.2	7190390
105.	Depth of guide channel, max (2)		.65%	3.2	7190398
106.	Width across nesting edge opening		.65%	3.2	Gage
107.	Height of tab or depth of undercut with respect to nesting edges, max (2)		.65%	3.2	Gage
108.	Distance from hole to outside edge of nesting area		.65%	3.2	Gage
109.	Distance from end of link to radhaz shield		.65%	3.2	2845938
	TESTS				
110.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.2
111.	Salt Spray	2/	2/	3.2	4.5.3
112.	Hardness	1/3/	.40%	3.2	4.5.4
113.	Weld Integrity		.65%	3.2	4.5.9

NOTES:
1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3
4/ To be performed with dummy cartridge assembled to link.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12910232
4.4.2.3	Loop, Long with Shield (Navy Use Only)				NEXT HIGHER ASSEMBLY 12910231
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR:					
201.	Height of stripper		1.50%	3.2	7190394
202.	Overall length of body, max		1.50%	3.2	Gage
203.	Depth of side detent, min (2)		1.50%	3.2	7190406
204.	Location of stripper opening		1.50%	3.2	Gage
205.	Improper or missing identification		1.50%	3.6	Visual
206.	Incorrect or missing protective finish		2.50%	3.2	Visual
207.	Poor workmanship		4.00%	3.8	Visual/6.4.1/6.4.2.2/ 6.4.3.2
NOTES:					

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Shield, Radhaz for Link M14A2N (Navy Use Only)	1 OF 1	12910233
			NEXT HIGHER ASSEMBLY 12910232

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:					
101.	Hardness	32	ACC-0 REJ-1	3.2	4.5.4
MINOR					
201.	Poor workmanship		4.00%	3.8	6.4.1/6.4.2.1/6.4.3.1/ Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Test Diagrams & Requirements for Link, Ctg Metallic Belt, 20mm: M14A2 or M14A2N		8447859
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:					
101.	Height from bottom of dummy cartridge to top of long loop, max	500	ACC-0 REJ-1	3.2	Gage 1/ 3/
102.	Distance from end of dummy cartridge to front guide of long loop, max.	500	ACC-0 REJ-1	3.2	Gage 1/ 3/
103.	Free hinging	800	ACC-0 REJ-1	3.4	4.5.7
104.	Cartridge extraction	500	ACC-5 REJ-6	3.2	8409005, 11838231, 11838232, 11838277 4.5.6.
105.	Pull and elongation	200	ACC-2 REJ-3	3.2	7190471/4.5.1 2/
106.	Tensile load	200	ACC-2 REJ-3	3.2	7274001, 4.5.5 2/
MINOR:	None defined.				

NOTES:

1/ Same samples to be used for inspection.
2/ Selected from cartridge extraction test sample.
3/ Major 101 and 102 inspections do not apply to the M14A2N link.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Link, Cartridge, Metallic Belt, 20MM: Packaging Data Sheet	1	1	P8447633/P12910231
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR:	None defined.				
MINOR:					
201.	Improper level of packaging and packing		1.50%	3.2	Visual
202.	Incorrect, missing or illegible markings on containers		1.50%	3.2	Visual
203.	Inadequate cleaning or drying of links		1.50%	3.2	Visual
204.	Unit container interior contaminated with dirt, lint, or other foreign matter		1.50%	3.2	Visual
205.	Improper orientation of links in unit container		1.50%	3.2	Visual
206.	Layers of links not separated properly		1.50%	3.2	Visual
207.	Improper closure of unit container		1.50%	3.2	Visual
208.	Improper packing of unit containers		1.50%	3.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

4.5.1 Pull and elongation. The link shall be assembled to the test fixture and subjected to the pull load specified on Dwg. 8447633 or 12910231. The distance, x-x, shall be measured and recorded. Observation shall be made for compliance with the applicable drawing requirement. The load shall then be increased at a uniform rate to that specified on Dwg. 8447859. The full load shall be removed and the pull load specified on Dwg. 8447633 or 12910231 reapplied. The distance x-x shall be remeasured and recorded. Observation shall be made for compliance with the permanent set requirement specified on the applicable drawing. In addition, observation shall be made for link fracture. Upon completion of the test, links so tested shall be scrapped.

4.5.2 Decarburization. The cross section of the sample short and long loops shall be polished, etched with 3 to 5% Nital solution and examined under a microscope of not less than 500 power. Observation shall be made for compliance with the applicable drawing requirement.

4.5.3 Salt spray. Salt spray testing shall be conducted in accordance with ASTM Method B117. Observation shall be made for compliance with the applicable requirement in DOD-P-16232.

4.5.4 Hardness. Hardness testing shall be conducted in accordance with ASTM Method E18. Observation shall be made for compliance with the applicable drawing requirement.

4.5.5 Tensile load. The sample links shall be assembled into belts of five (5) links each using three (3) dummy cartridges and the two (2) specified mandrels. The assembled belt shall be placed in the test fixture and subjected to the applicable tensile load. There shall be no belt separation. Upon removal of the load, observation shall be made for compliance with the applicable drawing required. Upon completion of the test, links so tested shall be scrapped.

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4.5.6 Cartridge extraction. The links shall be assembled to the test fixture and the load specified on the applicable drawing gradually applied. The extraction force shall be recorded. observation shall be made for compliance with the applicable drawing requirements.

4.5.7 Free hinging. The sample links shall be tested by assembling them into belts of 100 links each using M51 series 20MM dummy cartridges. Each belt shall be laid out horizontally to its full length with the open end of the links up. The end opposite the stop shown on Dwg. 8447859 shall be drawn back upon the remainder of the belt until the belt is completely reversed. This procedure shall then be repeated but the closed end of the links shall be up. observation shall be made for compliance with the applicable specification requirement.

4.5.8 Functioning. The sample of 600 M14A2 links shall be assembled into belts with 20MM cartridges and tested by the Government in any or all of the belt fed or linkless M61 gun systems (12 bursts of 50 rounds each) which are designed for the M14A2 link.

#4.5.8.1 The sample of 300 M14A2N linkes shall be assembled into belts with 20mm Cartridges and tested by the Government in the M197 Gun System, utilizing the M89E1 Feeder.

#4.5.9 Weld integrity. The welds on the sample links shall be tested in accordance with specification MIL-W-12332.

(NAVY USE ONLY)

5. PACKAGING

5.1 Preservation and packaging. The links shall be preserved and packaged for shipment in accordance with Packaging Data Sheet 8447633 or 12910231 with levels as specified in the contract (See 6.1).

5.2 Packing. The links shall be packed in accordance with Packaging Data Sheet 8447633 or 12910231 with levels as specified in the contract (See 6.1).

5.3 Marking. Marking shall be in accordance with Packaging Data Sheet 8447633 or 12910231.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, Data Item DI-R-1714 will apply.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM) or U.S. Army Armament Research and Development Center (ARDEC)drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.4 Definition of terms.

6.4.1 Deformed link. A link or radhaz shield whose shape or dimensions or both have been mutilated or altered by processing to the extent that it is unsuitable for its intended use.

6.4.2 Cracks - tears. See Figures 1, 2 and 3.

6.4.2.1 Cracks - tears in principal areas. Any visible fracture of the link extending through the stock thickness in depth, or a distance in length equal to or greater than the stock thickness in the areas specified in Figure 1 as depicted in Figure 2.

6.4.2.2 Cracks - tears in nonprincipal ares. Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length in all areas not specified in Figure 1 as depicted in Figure 3.

6.4.3 Burrs.

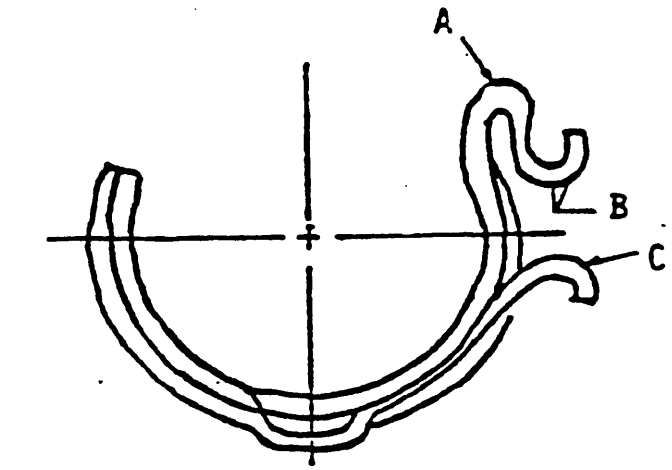
6.4.3.1 Burrs in principal areas. Any rough edge on any part of the link which, in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case or which would injure personnel in handling.

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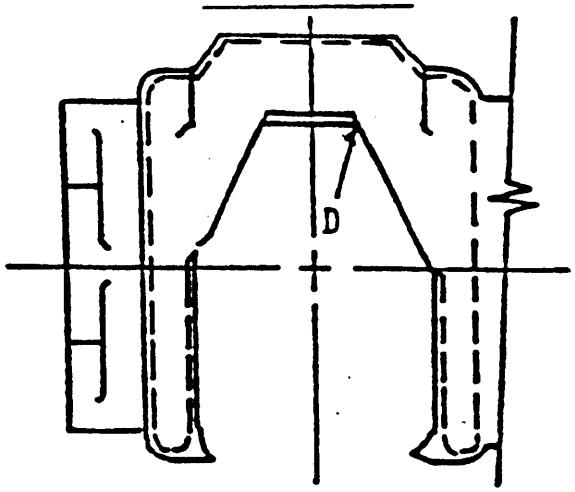
6.4.3.2 Burrs in nonprincipal areas. Any rough edge on any part of the link that does not fall within the description of 6.4.3.1.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

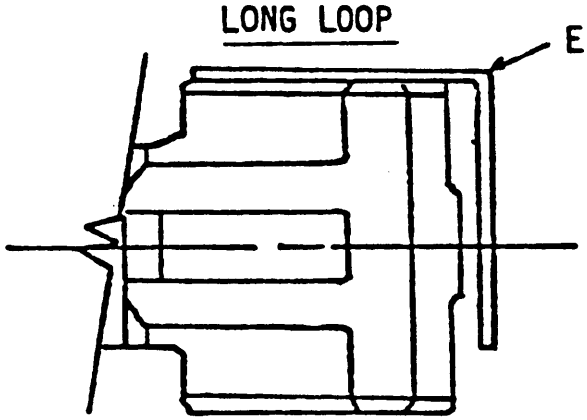
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SHORT LOOP



LONG LOOP

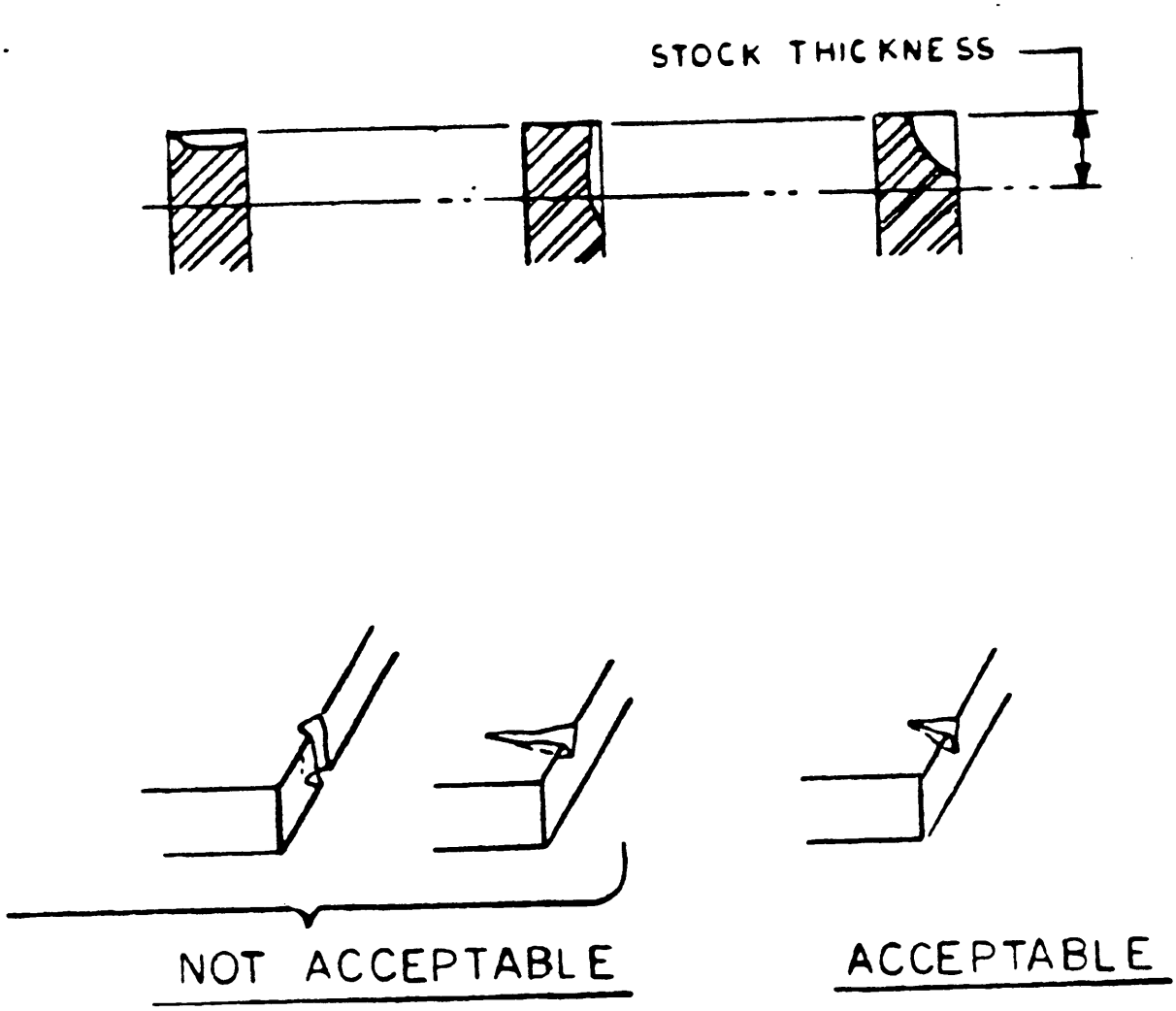


SHIELD, RADHAZ
(NAVY USE ONLY)

DESIGNATED PRINCIPAL AREAS

FIGURE 1

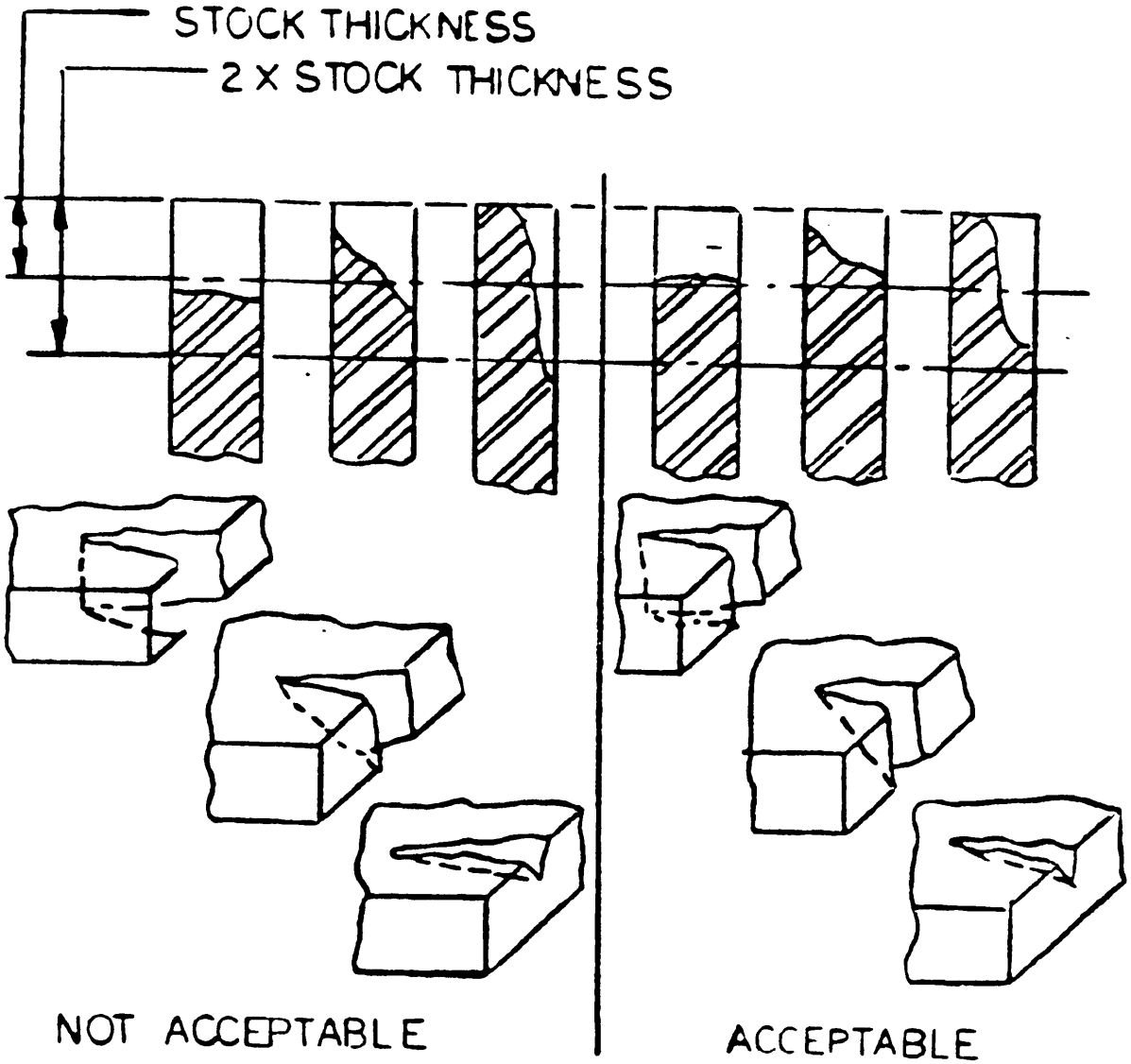
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CRACKS-TEARS IN PRINCIPAL AREAS

FIGURE 2

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CRACKS TEARS IN NONPRINCIPAL AREAS

FIGURE 3

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6.6 Subject term (key word) listing.

20mm M14A2 link
20mm M14A2N link
Radhaz shield
M61 20mm gun systems
M197 20mm gun systems
M89E1 feeder

Custodian:
Army-AR
Air Force-99

Preparing Activity:
Army-AR

(Project 1305-0932)

Review Activities:

Navy-OS

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-45194G	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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